
title: “Supply Chain Vector Analysis” author: “Guibril Ramde” format: pdf —

Tasks

Part 1 Inventory

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```
Warning: package 'tibble' was built under R version 4.5.2
```

```
Warning: package 'tidyr' was built under R version 4.5.2
```

```
Warning: package 'readr' was built under R version 4.5.2
```

```
Warning: package 'purrr' was built under R version 4.5.2
```

```
Warning: package 'dplyr' was built under R version 4.5.2
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.2.0      v readr      2.1.6
```

```
v forcats   1.0.1      v stringr    1.6.0
```

```
v ggplot2    4.0.2      v tibble     3.3.1
```

```
v lubridate  1.9.4      v tidyr      1.3.2
```

```
v purrr      1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
#Inventory data load
```

```
data = ("https://raw.githubusercontent.com/japhet125/Data\_Science\_605\_Hws/refs/heads/main/in
```

```
inventory_data = read.csv(data)
```

```
inventory_data
```

	Warehouse	Region1	Region2	Region3
1	A	1200	900	800
2	B	1000	1100	700
3	C	900	700	1000

Part 2 Demand

```
#Demand data load
data2 = ("https://raw.githubusercontent.com/japhet125/Data_Science_605_Hws/refs/heads/main/demand_data.csv")
demand_data2 = read.csv(data2)
demand_data2
```

	Region	Demand
1	Region1	2800
2	Region2	2400
3	Region3	2500

Tasks

Part A: Inventory Analysis

Using vector operations:

```
inventory_data$Region1
```

```
[1] 1200 1000 900
```

```
inventory_data$Region2
```

```
[1] 900 1100 700
```

```
inventory_data$Region3
```

```
[1] 800 700 1000
```

Calculate total inventory available in each region.

```
total_inventory <- c(  
  sum(inventory_data$Region1),  
  sum(inventory_data$Region2),  
  sum(inventory_data$Region3)  
)  
total_inventory
```

```
[1] 3100 2700 2500
```

Calculate inventory surplus or shortage relative to demand.

```
#If we have a small dataset  
region1 <- sum(inventory_data$Region1) - demand_data2$Demand[1]  
region1
```

```
[1] 300
```

```
region2 <- sum(inventory_data$Region2) - demand_data2$Demand[2]  
region2
```

```
[1] 300
```

```
region3 <- sum(inventory_data$Region3) - demand_data2$Demand[3]  
region3
```

```
[1] 0
```

```
#A large dataset we will use this R function for vector  
surplus_shortage <- total_inventory - demand_data2$Demand  
surplus_shortage
```

```
[1] 300 300 0
```

Identify which region is most at risk of stock shortages.

```
names(surplus_shortage) <- c("Region1", "Region2", "Region3")  
  
surplus_shortage
```

```
Region1 Region2 Region3  
      300      300       0
```

```
names(surplus_shortage)[which.min(surplus_shortage)]
```

```
[1] "Region3"
```

Interpret the business implications.

Region 3 has the greatest risk of an inventory risk of shortage because its available inventory is lowest relative to its demand. This could result in stock outs and an inability to fulfill customer orders. The company should consider reallocating inventory from regions with larger surpluses to Region 3 or increasing future inventory shipments to Region 3.

Part B: Shipment Matrix Analysis

Using matrix operations:

1. Create a shipment matrix from the data.

```
#load shipments data  
data3 <- "https://raw.githubusercontent.com/japhet125/Data_Science_605_Hws/refs/heads/main/sl  
shipment_data <- read_csv(data3)
```

```
Rows: 3 Columns: 4
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Warehouse
```

```
dbl (3): Region1, Region2, Region3
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
shipment_data
```

```
# A tibble: 3 x 4
  Warehouse Region1 Region2 Region3
  <chr>      <dbl>   <dbl>   <dbl>
1 A          300     400     500
2 B          400     350     250
3 C          500     300     200
```

```
shipment_matrix <- as.matrix(shipment_data[, 2:4])
shipment_matrix
```

```
      Region1 Region2 Region3
[1,]      300      400      500
[2,]      400      350      250
[3,]      500      300      200
```

1. Compute row totals and column totals.

```
#row total
warehouse_totals <- rowSums(shipment_matrix)
warehouse_totals
```

```
[1] 1200 1000 1000
```

```
#column total
region_total <- colSums(shipment_matrix)
region_total
```

```
Region1 Region2 Region3
      1200      1050      950
```

1. Determine which warehouse ships the most products.

```
names(warehouse_totals) <- c("A", "B", "C")
warehouse_totals
```

```
      A      B      C
1200 1000 1000
```

```
names(warehouse_totals)[which.max(warehouse_totals)]
```

```
[1] "A"
```

1. Determine which region receives the most products.

```
names(region_total) <- c("Region1", "Region2", "Region3")
region_total
```

```
Region1 Region2 Region3
      1200      1050      950
```

```
names(region_total)[which.max(region_total)]
```

```
[1] "Region1"
```

Visualization

```
# Visualization 1: Inventory vs Demand by Region
```

```
region_summary <- data.frame(
  Region = c("Region1", "Region2", "Region3"),
  Inventory = total_inventory,
  Demand = demand_data2$Demand
)
```

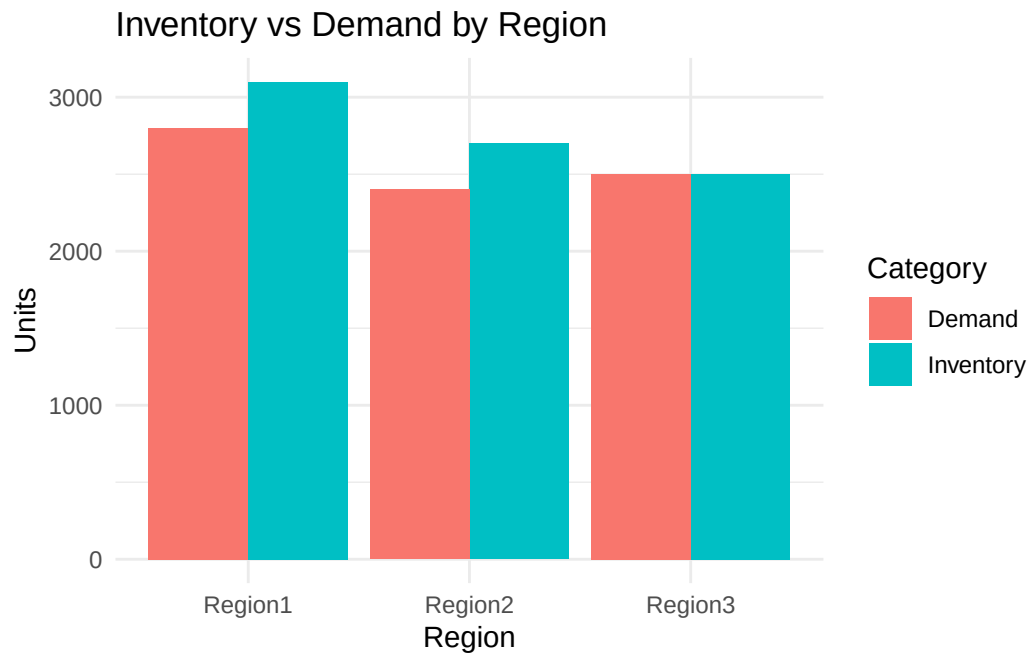
```
region_summary_long <- region_summary %>%
  pivot_longer(
    cols = c(Inventory, Demand),
    names_to = "Type",
    values_to = "Units"
  )
```

```
ggplot(region_summary_long, aes(x = Region, y = Units, fill = Type)) +
  geom_col(position = "dodge") +
  labs(
    title = "Inventory vs Demand by Region",
    x = "Region",
```

```

  y = "Units",
  fill = "Category"
) +
theme_minimal()

```



```

# Visualization 2: Total Shipments by Warehouse

warehouse_summary <- data.frame(
  Warehouse = c("A", "B", "C"),
  Total_Shipments = warehouse_totals
)

ggplot(warehouse_summary,
  aes(x = Warehouse, y = Total_Shipments)) +
  geom_col() +
  labs(
    title = "Total Shipments by Warehouse",
    x = "Warehouse",
    y = "Total Products Shipped"
  ) +
  theme_minimal()

```



Part C: Resource Allocation Model

Suppose inventory allocations must satisfy:

1. Express the system in matrix form.

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 18 \\ 26 \\ 22 \end{bmatrix}$$

Rewrite the system as $Ax=b$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$b = \begin{bmatrix} 18 \\ 26 \\ 22 \end{bmatrix}$$

2. Solve the System Computationally

```
#creating the coef matrix
A <- matrix(c(
  1,1,1,
  2,1,0,
  0,1,2

), nrow = 3, byrow = TRUE)

A
```

```
      [,1] [,2] [,3]
[1,]     1     1     1
[2,]     2     1     0
[3,]     0     1     2
```

```
#create b vector
b <- c(18,26,22)
b
```

```
[1] 18 26 22
```

```
#Solve for the unknown vector
#solution <- solve(A, b)
#solution
#det(A)
```

Solving x, y, z using R result to error due to the solution not having singular value but multiple values,

3. Verify the Solution

we have:

$$x+y+z=18$$

$$2x+y=26$$

$$y+2z=22$$

solution:

$$x+y+z=18$$

$$2x+y+0z=26$$

$$0x+y+2z=22$$

$$3x+3y+3z=66$$

$$3(x+y+z)=66 \implies (x+y+z)=66/3 \implies x+y+z=22$$

$x=22-y-z \implies$ replace x by its value, $22-y-z+y+z=18 \implies 22=18$ the system has no solution

we could also try $2x+y+y+2z=48 \implies 2x+2y+2z=48$ then $x+y+z=24$

$y=24-x-z$ so if we replace the values we get $x+24-x-z+z=18 \implies 24=18$

the system has no unique solution

recap: Since $(24 \neq 18)$, the equations are inconsistent. Therefore, there is no set of values for (x) , (y) , and (z) that satisfies all three equations.

This is also confirmed computationally in R because `solve(A, b)` returns a singular-system error and `det(A)` is equal to 0.

4. Business interpretation

Based on the analysis, the system of linear equations does not provide a valid number of truckloads that each warehouse should ship because the coefficient matrix is singular and the equations are inconsistent. Therefore, values for (x) , (y) , and (z) cannot be determined from the current system, and management would need to review the constraints before making a truckload allocation decision.

However, the results from Parts A and B still provide useful information for management. In Part B, Warehouse A ships the most products with 1,200 units, while Warehouses B and C each ship 1,000 units. This shows that Warehouse A currently contributes the most to the shipment activity. Part A also shows that Region 3 has no inventory surplus, while Regions 1 and 2 each have a surplus of 300 units. Therefore, Region 3 is more at risk if demand increases.

Mathematical models help management evaluate the needs of each region and the capacity of each warehouse so resources can be allocated efficiently. Systems of linear equations are useful in supply chain planning and logistics because they represent relationships between different constraints and help determine whether a proposed allocation is mathematically feasible.